# Machine Learning — Lecture 5

## Model Evaluation, Validation, and Selection

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# Lecture 5 at a Glance

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# Model Evaluation, Validation, and Selection

## Course Learning Outcome

**CLO5:** Evaluate machine learning models using appropriate metrics and justify decisions based on performance trade-offs and real-world context.

## Lecture purpose

- Move from “model output” to “model quality”.
- Evaluate regression and classification models correctly.
- Understand validation, cross-validation, and test discipline.
- Interpret **class imbalance**, thresholds, and error costs.
- Select models based on both metrics and application context.

## Why this lecture matters

A model is not useful only because it runs.

It is useful when we can answer:

- How well does it generalize?
- Which errors does it make?
- Which metric matters for the application?
- Which model should be selected?
- What is the cost of a wrong decision?

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💡 Main idea: evaluation is not only a calculation. It is a decision process that connects model behavior to the real-world problem.

# Learning Objectives

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# By the end of Lecture 5, you will be able to:

1. Distinguish training, validation, and test evaluation.
2. Explain why test data should not guide model selection.
3. Compute and interpret **regression metrics**: MAE, MSE, RMSE, and $R^2$.
4. Build and interpret a confusion matrix.
5. Compute accuracy, precision, recall, specificity, and $F_1$-score.
6. Explain why accuracy can be misleading under class imbalance.
7. Analyze threshold effects on false positives and false negatives.
8. Interpret ROC and precision-recall trade-offs conceptually.
9. Use cross-validation to support model selection.
10. Choose a model using performance metrics and real-world error costs.

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# Lecture Roadmap

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# From scores to responsible decisions

| Learning phase | Parts | Guiding question |
|---|---|---|
| **1. Establish evaluation discipline** | **I** | Why is training performance not enough, and what makes an evaluation trustworthy? |
| **2. Measure model performance** | **II–IV** | Which regression and classification metrics reveal the errors that matter? |
| **3. Make threshold and cost decisions** | **V** | How do decision thresholds change outcomes and practical consequences? |
| **4. Validate, select, and apply** | **VI–VIII** | How do we compare models fairly and justify a final choice in context? |

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💡 **Practice & consolidation:** compute metrics, inspect trade-offs, recognize evaluation mistakes, and defend a model choice.

# Part I

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# The Evaluation Problem

Building on Lectures 1–4, we now focus on **which evidence** supports a trustworthy evaluation and a defensible model choice.

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# Why Training Accuracy Is Not Enough

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# Memorization is not learning

A model can achieve **excellent training performance** by learning genuine patterns, but also by exploiting:

- noise and accidental correlations;
- duplicated examples;
- leaked information;
- sample-specific quirks.

## The key distinction

**Not sufficient**
Low training error

≠

**The real goal**
Low error on future unseen data

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💡 **Takeaway.** Evaluation on unseen data estimates **generalization**, rather than merely confirming that the model remembers its training examples.

# Development vs. Final Evaluation

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# Different data have different roles

| Dataset split | Main role | Should it guide model choice? |
|---|---|---|
| **Training** | fit parameters | yes, through learning |
| **Validation** | select model/hyperparameters | yes |
| **Test** | final unbiased estimate | no |

## Correct discipline

1. Train models on the training set.
2. Compare alternatives using the validation set.
3. Select the final model.
4. Evaluate once on the test set.

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If the test set influences repeated model choices, it becomes a validation set and no longer gives a clean final estimate.

# Data Leakage in Evaluation

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# When evaluation becomes too optimistic

**Data leakage** occurs when information from validation/test data enters training or model selection.

## Common forms

| Leakage source | Why it is problematic |
|---|---|
| Scaling before splitting | test distribution influences preprocessing |
| Feature selection using all data | test information affects chosen features |
| Repeated tuning on the test set | test set becomes part of development |
| Future variables used as features | prediction uses information unavailable at decision time |
| Duplicate examples across splits | model may see the same case during training |

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> A model affected by leakage may look excellent in evaluation and fail in real deployment.

# What Should Evaluation Answer?

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# Not one question, but several

A useful evaluation should answer:

1. How large are typical errors?
2. Are there rare but severe errors?
3. Which classes or groups are hardest?
4. Does the model generalize beyond training data?
5. Is performance stable across folds or splits?
6. Which error type is most costly?
7. Is the model good enough for the real decision?
8. Is it better than a baseline?

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> A 95% accurate model can still be useless if it misses the rare event that matters most.

# Baselines

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# Every model should beat a simple reference

A **baseline** is a simple reference model.

## Examples

| Task | Possible baseline |
|---|---|
| Regression | always predict the training mean |
| Binary classification | always predict the majority class |
| Multiclass classification | predict the most frequent class |
| Time-dependent data | predict the last observed value |
| Ranking | random or popularity-based ranking |

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> 💡 A complex model that barely beats a simple baseline may not justify its complexity.

# Part II

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# Regression Metrics

Regression evaluation measures how far numerical predictions are from actual values.

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# Regression Error Notation

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# Actual, prediction, residual

For each example $i$, compare the **actual value** with the model's **prediction**.

| Quantity | Notation | Meaning |
|---|---|---|
| **Actual value** | $y_i$ | Observed target |
| **Predicted value** | $\hat{y}_i$ | Model prediction |
| **Residual** | $e_i = y_i - \hat{y}_i$ | Signed difference |
| **Absolute error** | $\lvert e_i \rvert$ | Magnitude of the error |
| **Squared error** | $e_i^2$ | Penalizes larger errors more strongly |

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## Interpreting the residual

- $e_i > 0$: The model underpredicts.
- $e_i < 0$: The model overpredicts.
- $e_i = 0$: The prediction is exact.

💡 **Key idea.** Regression metrics summarize errors differently: **MAE** averages absolute errors, while **MSE** averages squared errors.

# Mean Absolute Error

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# Average size of the error

Mean Absolute Error is:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

## Interpretation

MAE is the average absolute **prediction error** in the same unit as the target.

## Strengths

- easy to explain;
- less sensitive to outliers than squared-error metrics;
- useful when error cost grows approximately linearly.

## Limitation

- does not strongly emphasize rare large mistakes.

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# Mean Squared Error and RMSE

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# Penalizing large errors

Mean Squared Error:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Root Mean Squared Error:

$$RMSE = \sqrt{MSE}$$

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## Interpretation

- MSE gives large errors much more influence.
- RMSE returns to the original unit of the target.

> 💡 If large mistakes are especially costly, **RMSE** may be more informative than MAE.

# Worked Example: Regression Metrics

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# Four predictions

| Example | Actual $y$ | Prediction $\hat{y}$ | Error $e$ | $\lvert e \rvert$ | $e^2$ |
|---|---|---|---|---|---|
| 1 | 100 | 90 | 10 | 10 | 100 |
| 2 | 120 | 130 | -10 | 10 | 100 |
| 3 | 80 | 85 | -5 | 5 | 25 |
| 4 | 200 | 160 | 40 | 40 | 1600 |

Then:

$$MAE = \frac{10 + 10 + 5 + 40}{4} = 16.25.$$

$$MSE = \frac{100 + 100 + 25 + 1600}{4} = 456.25.$$

$$RMSE = \sqrt{456.25} \approx 21.36.$$

The large error of 40 strongly increases MSE and **RMSE**.

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# Coefficient of Determination

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# $R^2$

The coefficient of determination compares prediction error with a mean-based baseline:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

- **SSE (Sum of Squared Errors):** $\sum_i (y_i - \hat{y}_i)^2$ — squared prediction errors.
- **SST (Total Sum of Squares):** $\sum_i (y_i - \bar{y})^2$ — total variation around the actual mean $\bar{y}$.

## Interpretation

| Value | Meaning |
|---|---|
| $1$ | perfect predictions on evaluated data |
| $0$ | same squared error as predicting the evaluated sample mean |
| $< 0$ | worse than that mean-based prediction |
| close to $1$ | much less squared error than the mean baseline |

A high training $R^2$ can still indicate overfitting if validation/test performance is weak.

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# Worked Example: Interpreting $R^2$

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# Use the same four regression predictions

From the previous example, $y = (100, 120, 80, 200)$ and $\hat{y} = (90, 130, 85, 160)$.

Mean actual value: $\bar{y} = 125$

$$SSE = 100 + 100 + 25 + 1600 = 1825$$

$$SST = (100 - 125)^2 + (120 - 125)^2 + (80 - 125)^2 + (200 - 125)^2 = 8300$$

$$R^2 = 1 - \frac{1825}{8300} \approx 0.780$$

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> The predictions reduce squared error by about **78%** relative to a constant prediction using the evaluated sample mean. This does **not** mean that 78% of future individual predictions are correct.

# Choosing a Regression Metric

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# Match metric to context

| Application | Metric emphasis | Reason |
|---|---|---|
| Rent prediction | MAE | easy to interpret in currency |
| Medical dosage | RMSE or safety-specific metric | large errors may be dangerous |
| Energy forecasting | MAE + RMSE | typical and extreme errors both matter |
| Scientific model fit | $R^2$ plus residual analysis | explains variation but not all error behavior |
| Business forecasting | MAE, RMSE, percentage error | cost interpretation matters |

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💡 There is no universally best metric. The metric should match the consequence of **prediction error**.

# Residual Analysis

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# Metrics are not the whole story

**Residuals** ($e_i = y_i - \hat{y}_i$) reveal patterns that aggregate regression metrics may hide.

## Systematic patterns

- **Underprediction:** Residuals are consistently positive.
- **Overprediction:** Residuals are consistently negative.
- **Nonlinear structure:** Residuals follow a curved pattern.

## Error variability

- **Increasing spread:** Error variance changes across predictions.
- **Outliers:** A few examples have unusually large residuals.
- **Group differences:** Errors are concentrated in particular groups.

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## Example

If a model consistently **underpredicts expensive apartments**, it may fail to capture pricing patterns in the high-end market, even when its overall MAE is low.

> 💡 **Key idea.** Regression metrics summarize **how much** error occurs; residual analysis reveals **where and how** the model makes errors.

# Visualize the Regression Errors

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# One large residual changes RMSE strongly




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💡 The fourth prediction has the largest **residual** ($+40$). Squaring its error contributes **1600 of the total 1825 squared-error units**.

# Part III

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# Classification Metrics

Classification evaluation begins by counting the types of correct and incorrect decisions.

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# Confusion Matrix

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# Four outcomes for binary classification

Assume class $1$ is the positive class.

| | Predicted Positive | Predicted Negative |
|---|---|---|
| **Actual Positive** | True Positive (TP) | False Negative (FN) |
| **Actual Negative** | False Positive (FP) | True Negative (TN) |

## Meaning

| Term | Interpretation |
|---|---|
| TP | positive case correctly detected |
| TN | negative case correctly rejected |
| FP | false alarm |
| FN | missed positive case |

> In fraud detection, FN means fraud was missed; FP means a legitimate transaction was flagged.

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# Visual Confusion Matrix

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# Connect the four cells to the metrics

## 200 labeled examples · positive class = 1

| Actual | Predicted: Positive | Predicted: Negative |
|---|---|---|
| positive | **TP = 40** correct detection | **FN = 20** missed positive |
| negative | **FP = 10** false alarm | **TN = 130** correct rejection |

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💡 In this example, **recall = 40/60** asks how many actual positives were detected, while **precision = 40/50** asks how reliable positive alerts were.

# Accuracy

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# Overall fraction correct

Accuracy is:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

## Useful when

- classes are balanced;
- all error types have similar cost;
- the task is not dominated by a rare positive class.

## Misleading when

- one class is much more common;
- **false positives** and false negatives have different costs;
- the model ignores the minority class.

**Accuracy** answers “How often is the model correct?” but not “Which mistakes does it make?”

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# Precision

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# Of predicted positives, how many were correct?

Precision is:

$$Precision = \frac{TP}{TP + FP}$$

## High precision means

When the model predicts positive, it is usually correct.

## Important when

False positives are costly.

## Examples:

- falsely accusing a transaction of fraud;
- sending unnecessary medical alarms;
- blocking legitimate users;
- labeling normal activity as suspicious.

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**Precision** focuses on the reliability of positive predictions.

# Recall

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# Of actual positives, how many were found?

Recall is:

$$Recall = \frac{TP}{TP + FN}$$

**Recall** is also called sensitivity or true positive rate.

## High recall means

The model finds most of the actual positive cases.

## Important when

False negatives are costly.

## Examples:

- missed disease;
- missed fraud;
- failing to identify at-risk students;
- missed safety alert.

> Recall focuses on not missing positive cases.

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# Specificity

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# Of actual negatives, how many were rejected?

Specificity is:

$$Specificity = \frac{TN}{TN + FP}$$

**Specificity** is the true negative rate.

## High specificity means

The model correctly rejects most negative cases.

## Important when

False positives cause burden, cost, or harm.

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> 💡 Recall and specificity often move in opposite directions when the **decision threshold** changes.

# F1-Score

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# Balancing precision and recall

The $F_1$-score is the harmonic mean of precision and recall:

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

## Useful when

- positive class is important;
- class imbalance exists;
- both FP and FN matter;
- a single summary is needed.

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## Caution

$F_1$ ignores **true negatives**. When no positive predictions are made, precision needs an explicit zero-division convention.

> A high $F_1$ means the model has a reasonable balance between finding positives and avoiding **false positives**.

# Worked Example: Confusion Matrix

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# Calculate metrics

**Given:** $TP = 40, \quad FP = 10, \quad FN = 20, \quad TN = 130$

| Metric | Calculation |
|---|---|
| **Accuracy** | $\frac{40 + 130}{40 + 10 + 20 + 130} = \frac{170}{200} = 0.850$ |
| **Precision** | $\frac{40}{40 + 10} = 0.800$ |
| **Recall** | $\frac{40}{40 + 20} \approx 0.667$ |
| **Specificity** | $\frac{130}{130 + 10} \approx 0.929$ |
| $F_1$ **score** | $2 \times \frac{0.80 \times 0.667}{0.80 + 0.667} \approx 0.727$ |

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# Which Metric Matters?

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# It depends on the error cost

| Context | More costly error | Metric emphasis |
|---|---|---|
| Cancer screening | false negative | recall/sensitivity |
| Spam filtering | false positive may lose important email | precision |
| Fraud detection | false negative often expensive | recall + cost analysis |
| Search results | false positives reduce quality | precision |
| Safety monitoring | missed danger is severe | recall |
| Manual review queue | too many false alarms overload reviewers | precision + workload |

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💡 **Metric** choice is part of problem formulation.

# Part IV

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# Class Imbalance

When one class dominates, accuracy can hide failure on the important minority class.

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# Majority-Class Baseline

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# A model can be accurate and useless

Suppose 1,000 transactions contain:

- 990 legitimate transactions;
- 10 fraudulent transactions.

A classifier that predicts every transaction as legitimate obtains:

$$Accuracy = \frac{990}{1000} = 99\%$$

But:

$$Recall_{\text{fraud}} = 0$$

## Why this is bad

The model detects no fraud.

> Always compare against a **baseline** and inspect class-specific performance.

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# Imbalance Changes Interpretation

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# High accuracy may not be impressive

Suppose **98% of cases are negative**. A model that always predicts **negative** achieves **98% accuracy**, but **0% recall** for the positive class.

## What should we evaluate?

### Classification performance

- Confusion matrix;
- Precision;
- Recall;
- $F_1$ score.

### Errors and decisions

- Class-specific errors;
- Decision-threshold behavior;
- Costs of FP and FN;
- Precision–recall curve, when appropriate.

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## Examples of imbalanced tasks

**Fraud detection · Rare disease diagnosis · Equipment failure prediction · Intrusion detection · At-risk student identification**

**Key idea.** When the positive class is rare but important, evaluate **precision, recall, and the costs of false positives and false negatives**. High accuracy alone may hide poor detection of the cases that matter.

# Balanced Accuracy

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# Average of class-level correctness

Balanced accuracy averages sensitivity and **specificity**:

$$Balanced\ Accuracy = \frac{Recall + Specificity}{2}$$

This helps when classes are imbalanced.

## Example

If:

$$Recall = 0.60, \qquad Specificity = 0.95$$

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then:

$$Balanced\ Accuracy = \frac{0.60 + 0.95}{2} = 0.775$$

💡 Balanced **accuracy** gives both classes a more equal role than ordinary accuracy.

# Part V

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# Thresholds and Trade-offs

Many classifiers output probabilities or scores. The threshold controls the decision.

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# Threshold Movement

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# Lower or higher threshold?

If a classifier predicts probability $p = P(y = 1 \mid x)$: $\hat{y} = 1 \quad \text{if} \quad p \geq t$

## Lower threshold

- predicts more positives;
- tends to increase recall;
- tends to increase **false positives**;
- may reduce precision.

## Higher threshold

- predicts fewer positives;
- may improve precision (not guaranteed);
- can increase **false negatives**;
- cannot increase recall when scores and labels are fixed.

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💡 Thresholds are chosen based on the cost of decisions, not only by convention.

# Threshold Example

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# Predicted fraud probabilities

| Transaction | True class | $P(\text{fraud})$ |
|---|---|---|
| A | Fraud | 0.91 |
| B | Legitimate | 0.64 |
| C | Fraud | 0.48 |
| D | Legitimate | 0.22 |
| E | Fraud | 0.18 |
| F | Legitimate | 0.07 |

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**At threshold $t = 0.5$**

Predicted positives: $A,\ B$

**At threshold $t = 0.2$**

Predicted positives: $A,\ B,\ C,\ D$

> Lowering the **threshold** detects C but also flags D.

# Visual Threshold Comparison

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# Changing the threshold changes which cases are flagged

## Same probabilities · different decision thresholds

| Transaction | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| True class | Fraud | Legit. | Fraud | Legit. | Fraud | Legit. |
| Fraud score | 0.91 | 0.64 | 0.48 | 0.22 | 0.18 | 0.07 |
| t = 0.50 | Flag | Flag | Pass | Pass | Pass | Pass |
| t = 0.20 | Flag | Flag | Flag | Flag | Pass | Pass |

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💡 At $t = 0.50$, only A and B are flagged. At $t = 0.20$, C and D join them: **one additional true positive and one additional false positive**.

# Worked Threshold Metrics

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# Compute precision and recall at two thresholds

Using the six labeled transactions A–F shown earlier:

| Threshold | TP | FP | FN | TN | Precision | Recall |
|---|---|---|---|---|---|---|
| $t = 0.50$ | 1 | 1 | 2 | 2 | $1/2 = 0.50$ | $1/3 \approx 0.33$ |
| $t = 0.20$ | 2 | 2 | 1 | 1 | $2/4 = 0.50$ | $2/3 \approx 0.67$ |

Lowering the **threshold** raises recall, **but precision stays the same in this example**.

> This is why we say precision *may* decrease as the threshold drops: it is **not guaranteed** to change in only one direction.

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# Precision–Recall Trade-off

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# How the decision threshold changes predictions

The **decision threshold** determines which predicted probabilities are classified as positive.

## Lower threshold

- **More** examples predicted positive.
- More actual positives may be detected.
- **Recall cannot decrease** for fixed scores.
- False positives may increase.
- Precision may **increase or decrease**.

## Higher threshold

- **Fewer** examples predicted positive.
- Predictions become more selective.
- **Recall cannot increase** for fixed scores.
- Some actual positives may be missed.
- Precision may **increase or decrease**.

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💡 **Key idea.** Lowering the threshold generally favors **recall**, while raising it makes positive predictions more selective–but **precision is not guaranteed to improve**. Choose the threshold using validation data and the costs of FP and FN.

# ROC Curve

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# True positive rate vs. false positive rate

The **ROC curve** shows how two rates change as the classification threshold varies:

$$\mathrm{TPR} = \frac{TP}{TP + FN}, \qquad \mathrm{FPR} = \frac{FP}{FP + TN}$$

- **True positive rate (TPR):** Proportion of actual positives correctly detected; also called **recall**.
- **False positive rate (FPR):** Proportion of actual negatives incorrectly classified as positive.

## Interpreting ROC AUC

**ROC AUC** summarizes how well a model ranks positive examples above negative examples.

- **AUC near 1:** Strong ranking ability.
- **AUC near 0.5:** Similar to random ranking.
- **Higher AUC:** Generally better discrimination between classes.

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**Class imbalance matters.** ROC AUC may look favorable even when false alarms are operationally costly. For rare positive classes, also inspect **precision–recall performance**.

# Precision–Recall Curve

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# Often more informative for rare positives

The **precision–recall curve** shows the relationship between:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}$$

as the **decision threshold** changes.

## Why is it useful?

- **Precision:** Of all flagged examples, how many are actually positive?
- **Recall:** Of all actual positives, how many are detected?

Precision–recall analysis is especially useful when:

- the positive class is **rare**;
- false positives and false negatives have different costs;
- detecting the positive class is the main objective.

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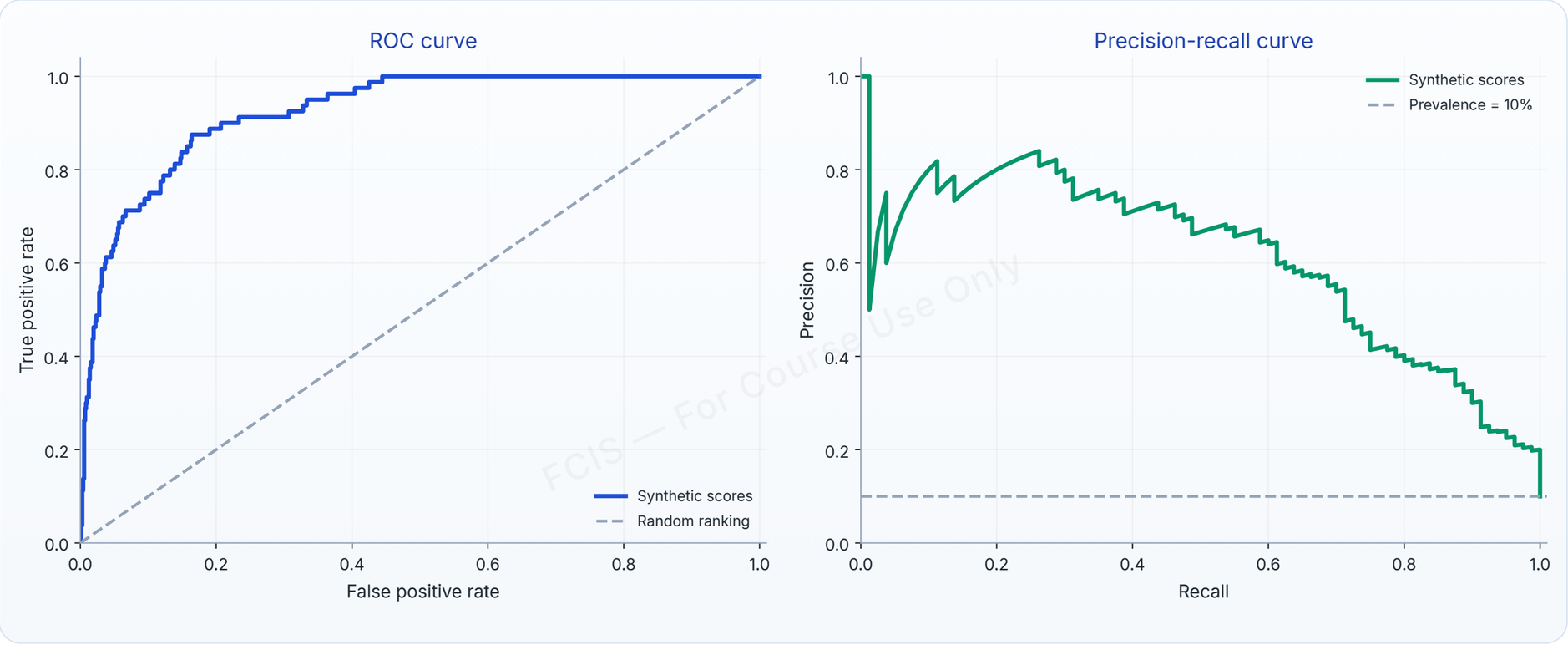
💡 **Key idea.** For tasks such as **fraud detection** or **rare disease diagnosis**, precision–recall analysis helps assess whether a model detects important cases without producing too many false alarms.

# ROC and Precision–Recall Curves

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# Same scores, different perspectives

A **synthetic dataset** contains 80 positives and 720 negatives (**10% prevalence**). Both curves are constructed from exactly the same scores as the decision threshold varies.




💡 **Key distinction.** ROC compares detection rate with the **false positive rate**. Precision–recall shows how many flagged examples are truly positive, making it especially useful when positives are rare.

# Cost-Sensitive Evaluation

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# Different errors have different consequences

In some applications, **false positives (FP)** and **false negatives (FN)** have very different costs.

Suppose:

- **False negative:** 500 SAR per missed positive case.
- **False positive:** 10 SAR per false alarm.

## Total error cost

$$\boxed{Cost = 500(FN) + 10(FP)}$$

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A model with **lower accuracy** may still be preferable if it avoids expensive false negatives.

> 💡 **Key idea.** Model evaluation should reflect the **consequences of errors**, not just how many predictions are correct.

# Worked Cost Example

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# Two fraud models, same validation set

Both models are evaluated on **500 transactions**: 50 fraudulent and 450 legitimate.

| Model | TP | FP | FN | TN | Accuracy |
|---|---|---|---|---|---|
| **A** | 40 | 20 | 10 | 430 | **94.0%** |
| **B** | 47 | 60 | 3 | 390 | **87.4%** |

Assume **500 SAR per FN** and **10 SAR per FP**.

## Compare total error costs

$$Cost_A = 10(500) + 20(10) = \mathbf{5{,}200}\ \text{SAR}$$

$$Cost_B = 3(500) + 60(10) = \mathbf{2{,}100}\ \text{SAR}$$

**Decision:** Model B has lower accuracy, but its estimated error cost is **3,100 SAR lower** because it misses fewer fraud cases.

💡 **Lesson.** The preferred model depends on the **costs of FP and FN**. Different cost assumptions could lead to a different choice.

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# Part VI

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# Validation and Model Selection

Building on the train/validation/test discipline from Lecture 1, we now **compare real candidates** and choose algorithms, features, and hyperparameters from validation evidence.

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# What Is a Hyperparameter?

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# Chosen before or outside training

**Parameters** (e.g., coefficients and weights) are **learned from training data**.

**Hyperparameters** control model structure or training and are chosen outside an individual training fit.

| Model | Examples of hyperparameters |
|---|---|
| Ridge / Lasso | Regularization strength $\lambda$ |
| Logistic regression | Regularization strength |
| KNN | Number of neighbors $k$ |
| Decision tree | Maximum depth, minimum samples per leaf |
| Neural network | Learning rate, number of layers, hidden units |
| K-Means | Number of clusters $K$ |

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**Selection matters.** Tune supervised-model hyperparameters using **validation data or cross-validation**, never the final test set. A **classification threshold** is a separate decision choice: changing it alters predicted labels without retraining the model.

# Train–Validation–Test Workflow

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# Development first, final evaluation last

A reliable workflow separates **training**, **model selection**, and **final evaluation**.

### 1. Prepare & Train

- Split into train, validation, and test sets.
- Fit preprocessing on training data only.
- Train candidate models.

### 2. Validate & Select

- Compare models on validation data.
- Tune hyperparameters and, if needed, thresholds.
- Select the final model.

### 3. Finalize & Test

- Refit on train + validation if appropriate.
- Refit preprocessing with the model.
- Evaluate once on the untouched test set.

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💡 **Key idea.** Training data is used to learn the model, validation data guides model selection, and the **untouched test set** estimates performance on unseen data.

# Cross-Validation

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# More reliable comparison when data are limited

**$k$-fold cross-validation** evaluates a model across multiple training–validation splits.

1. Divide the **development data** into $k$ folds.
2. Fit preprocessing and train the model on $k-1$ **folds**.
3. Evaluate on the **remaining validation fold**.
4. Repeat until every fold has served as validation, then **average the scores**.

## Why use it?

Each example is used for **validation once** and for **training $k-1$ times**, making evaluation less dependent on one particular split.

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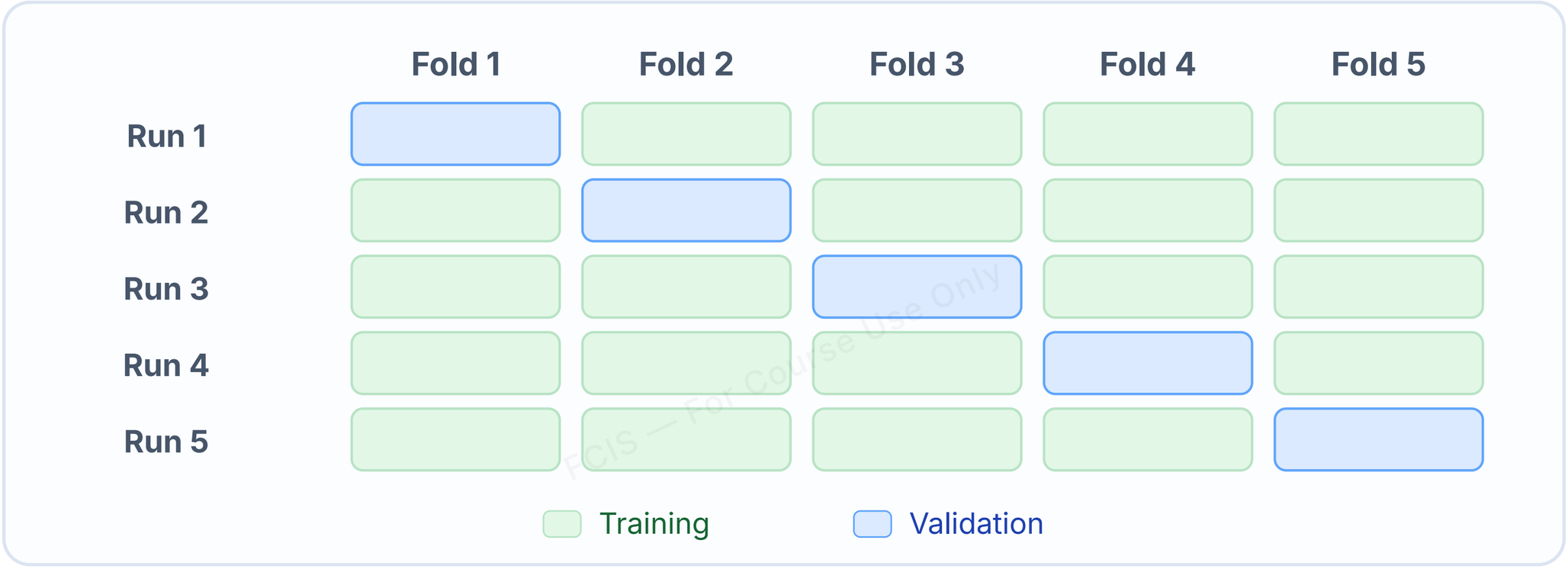
💡 **Key idea.** Use cross-validation to compare models and tune hyperparameters while keeping the **final test set untouched**. Choose stratified, group-aware, or time-aware folds when appropriate.

# Cross-Validation at a Glance

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# Five rounds, five different validation folds

In **5-fold cross-validation**, each run trains on four folds and validates on the fifth.




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**Avoid leakage:** Fit preprocessing separately on the training folds of each run. Never use the held-out fold to calculate scaling statistics or other learned transformations.

# Cross-Validation Score

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# Combine the results across folds

Suppose a KNN model with $k = 7$ **neighbors** produces these illustrative validation accuracies:

| Validation fold | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Accuracy** | 0.89 | 0.93 | 0.90 | 0.92 | 0.91 |

## Average validation performance

$$\overline{\text{Accuracy}}_{\text{CV}} = \frac{0.89 + 0.93 + 0.90 + 0.92 + 0.91}{5} = \boxed{0.91}$$

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**Interpretation.** The mean cross-validation accuracy is **91%**. We report the average rather than selecting the best fold's score of 93%.

💡 Cross-validation produces **multiple validation scores**. Their mean summarizes performance, while their variation reveals sensitivity to the data split.

# Model Selection Example

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# Choosing $k$ for KNN

Consider the following **illustrative 5-fold cross-validation results**.

| $k$ | Mean training accuracy | Mean CV accuracy |
|---|---|---|
| 1 | **1.00** | 0.84 |
| 3 | 0.93 | 0.89 |
| 7 | 0.88 | **0.91** |
| 15 | 0.82 | 0.86 |

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## Questions

1. Which $k$ achieves the highest training accuracy?
2. Which $k$ achieves the highest mean CV accuracy?
3. Which $k$ should we select, and why?
4. What does the performance gap for $k = 1$ suggest?

# Choosing $k$ for KNN

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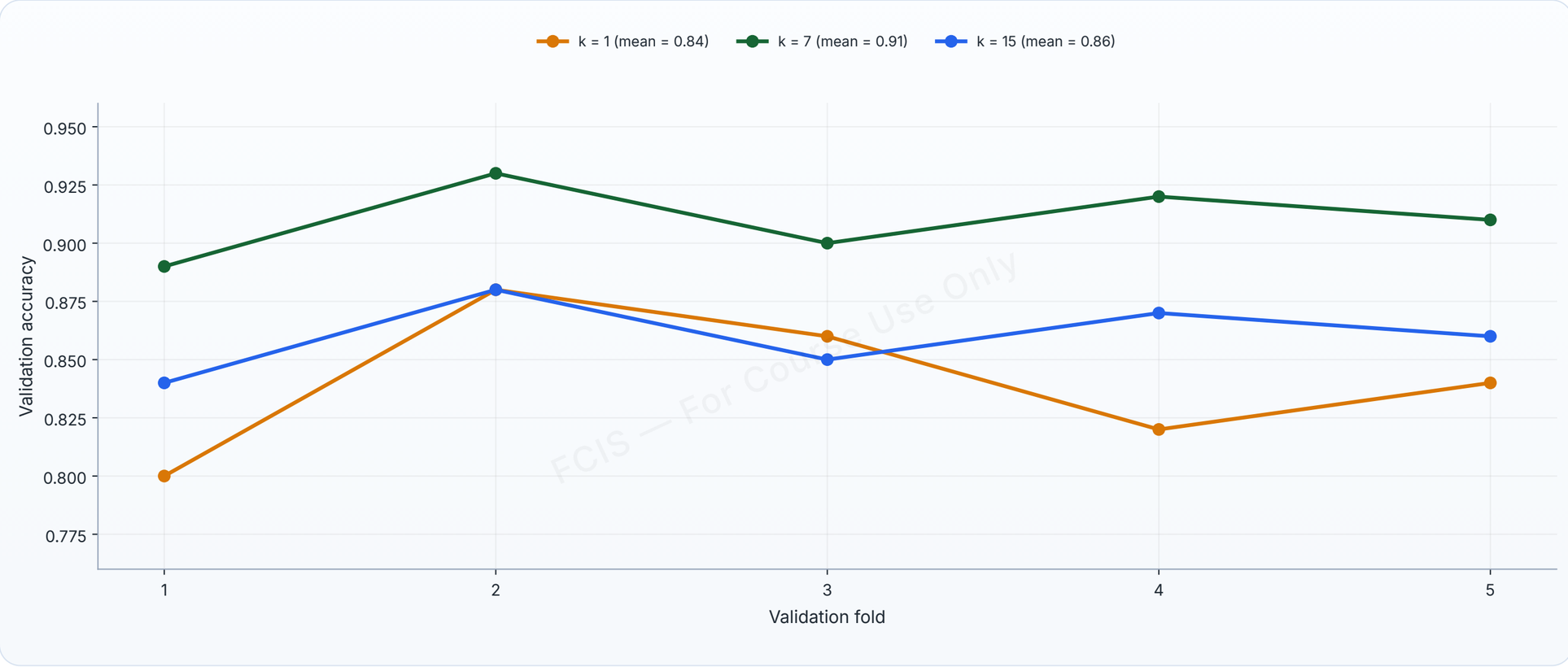
**Solution:** Select $k = 7$ because it achieves the highest mean CV accuracy (**91%**). For $k = 1$, perfect training accuracy but lower CV accuracy (**84%**) suggests **overfitting**.

# Cross-Validation Variability

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# A good mean can hide variation across folds

Illustrative five-fold validation accuracies for **three KNN settings** from the previous slide. These are example values, not measured experimental results.




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💡 **Key idea.** Compare both **mean validation performance** and **fold-to-fold variability**. Use cross-validation to select hyperparameters, then evaluate the finalized model on the untouched test set.

# Nested Decision Discipline

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# Avoid selecting from the test set

## Incorrect workflow

**Try many models**
Compare candidates

→

**Choose best test score**
Test set influences selection

→

**Report same test score**
Optimistically biased result

## Correct workflow

**Try many models**
Compare candidates

→

**Choose using validation**
Select the model

→

**Evaluate on test set**
Report final performance once

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## Why?

Repeatedly using the test set to select models makes the reported performance **too optimistic**.

> **Takeaway.** Use validation for model selection; reserve the **test set** for final evaluation.

# Part VII

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# Worked Evaluation Case

Now we combine metrics, thresholds, and context in one classification example.

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# Case: At-Risk Student Detection

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# Problem setting

A university wants to identify students who may fail a course early enough to offer support.

Positive class:

$$y = 1 \quad \text{means at risk.}$$

Possible errors:

| Error | Meaning | Consequence |
|---|---|---|
| False positive | student flagged but not actually at risk | unnecessary advising workload |
| False negative | at-risk student not flagged | missed opportunity for support |

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💡 In this context, false negatives may be more serious than **false positives**.

# Case: Two Models

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# Validation results

| Model | TP | FP | FN | TN |
|---|---|---|---|---|
| A | 42 | 18 | 28 | 212 |
| B | 55 | 45 | 15 | 185 |

## Compute

For each model:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

## Guiding question

Which model is more suitable if the main goal is to avoid missing at-risk students?

# Case: Interpretation

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# Metrics are connected to action

| Metric | Model A | Model B |
|---|---|---|
| **Precision** | $\frac{42}{42+18} = 0.700$ | $\frac{55}{55+45} = 0.550$ |
| **Recall** | $\frac{42}{42+28} = 0.600$ | $\frac{55}{55+15} \approx 0.786$ |
| $F_1$ **score** | $\frac{2(42)}{2(42)+18+28} \approx 0.646$ | $\frac{2(55)}{2(55)+45+15} \approx 0.647$ |
| **Students flagged** | **60** | **100** |

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## Interpretation

- **Model A:** More precise alerts and fewer students flagged.
- **Model B:** Identifies more at-risk students (**55 vs. 42**) but flags **40 additional students**.
- Both models have nearly identical $F_1$ **scores**, despite different precision–recall trade-offs.

**Decision.** If avoiding missed at-risk students is the priority, **Model B** is preferable: it misses only **15** students instead of **28**, provided the additional advising workload is acceptable.

# Case: Add Cost

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# Make the decision explicit

Suppose:

- **false positive** cost = 1 advising unit;
- **false negative** cost = 5 missed-support units.

Model A:

$$Cost_A = 18(1) + 28(5) = 158$$

Model B:

$$Cost_B = 45(1) + 15(5) = 120$$

## Decision

Model B has lower total cost under these assumptions.

> 💡 A model should be selected using the consequences of errors, not metric labels alone.

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# Part VIII

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# Common Evaluation Mistakes

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# When good scores can be misleading

A high score does not always mean a good model.

**Trustworthy evaluation requires the right data, metrics, and context.**

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# Mistake 1

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# Evaluating on training data only

A student reports **99% training accuracy** and concludes that the model is excellent.

## Why is this misleading?

Training accuracy measures performance on data the model has **already seen**. A high score may result from:

- memorizing training examples;
- fitting noise instead of general patterns;
- excessive model complexity;
- information leakage.

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A model can perform very well during training but poorly on **unseen data**.

> **Lesson.** Training accuracy alone does not demonstrate **generalization**. Evaluate using properly separated validation and test data.

# Mistake 2

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# Optimizing directly on the test set

A student repeatedly changes the:

- model type;
- selected features;
- hyperparameters;
- classification threshold;

until the **test score** becomes high.

## Why is this a problem?

The test set is now influencing model selection. Repeated experimentation can lead to **overfitting the test set**, even without directly training on it.

> **Lesson.** Use **validation data or cross-validation** for model development. Reserve the **untouched test set** for final evaluation.

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# Mistake 3

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# Using the wrong metric

A rare-disease classifier achieves **98% accuracy**.

But suppose only **2% of patients** have the disease.

A model that predicts **“healthy” for everyone** also achieves:

- **Accuracy:** 98%
- **Recall for disease cases:** 0%
- **Detected disease cases:** None

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## Better evaluation

Examine the **confusion matrix, precision, recall, false-negative costs**, and decision-threshold behavior.

> 💡 **Lesson.** A metric is useful only when it reflects the **goal and consequences** of the prediction task. High accuracy alone may hide critical failures.

# Mistake 4

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# Ignoring deployment context

Even a model with excellent validation results may be unsuitable in practice.

## Questions to consider

- **Feature availability:** Will inputs be available at prediction time?
- **Decision costs:** Are predictions worth acting on?
- **Group performance:** Are errors distributed unequally?
- **Speed:** Can the model meet time constraints?
- **Explainability:** Are understandable decisions required?
- **Data shift:** Will future data resemble the evaluation data?

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💡 **Lesson.** A useful model must perform well **and** satisfy the requirements of its real-world application.

# Practice & Consolidation

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# Evaluation Decisions and Common Mistakes

**How do we know whether a model is truly good?**

Use the right metrics, validation strategy, and decision threshold—and consider the consequences of errors.

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# Activity: Choose the Metric

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# Match evaluation metrics to the task

For each application, choose suitable **evaluation metrics** and explain why they matter.

| Task | Relevant metric(s) | Why? |
|---|---|---|
| Predict apartment rent | | |
| Detect rare fraud | | |
| Identify at-risk students | | |
| Classify product category | | |
| Forecast energy demand | | |
| Medical screening | | |

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| Identify at-risk students | | |
| Classify product category | | |
| Forecast energy demand | | |
| Medical screening | | |

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**Possible answers:**

- **Rent / Energy:** MAE or RMSE to measure prediction error.
- **Fraud:** Precision, recall, and error costs.
- **At-risk students:** Recall, precision, and intervention workload.
- **Product category:** Accuracy, confusion matrix; macro-$F_1$ if classes are imbalanced.
- **Medical screening:** Sensitivity (recall), specificity, and error costs.

**Remember:** There may be more than one suitable metric.

# Activity: Interpret a Confusion Matrix

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# Calculate and explain the metrics

A binary classifier produces:

$$TP = 30, \quad FP = 15, \quad FN = 10, \quad TN = 145$$

## Calculate

1. **Accuracy**
2. **Precision**
3. **Recall (sensitivity)**
4. **Specificity**
5. $F_1$ **score**

## Interpret

> Is the model better at correctly rejecting **negative cases** or detecting **positive cases**? Compare specificity with recall.

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# Calculate and explain the metrics

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## Calculate

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## Interpret

Is the model better at correctly rejecting **negative cases** or detecting **positive cases**? Compare specificity with recall.

**Solution:**

- Accuracy = **87.5%**; Precision = **66.7%**; Recall = **75.0%**
- Specificity = **90.6%**; $F_1$ = **70.6%**

**Interpretation:** The model correctly rejects a greater proportion of negatives (**90.6%**) than the proportion of positives it detects (**75%**).

# Activity: Threshold Reasoning

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# Predict the effect before computing

A binary classifier changes its decision threshold from **0.5 to 0.3**, while keeping the predicted scores fixed.

Predict the effect of this change:

| Quantity | Increase, decrease, or uncertain? |
|---|---|
| Number of predicted positives | |
| Recall | |
| False positives (FP) | |
| Precision | |
| False negatives (FN) | |

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|---|---|
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| Recall | |
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| Precision | |
| False negatives (FN) | |

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**Solution:**

- **Predicted positives:** Increase or remain unchanged.
- **Recall:** Increase or remain unchanged.
- **False positives:** Increase or remain unchanged.
- **Precision:** May increase or decrease (or remain unchanged).
- **False negatives:** Decrease or remain unchanged.

**Why?** A lower threshold flags more examples, but not all newly flagged examples are truly positive.

# Activity: Select a Model

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# Validation evidence and practical constraints

Three illustrative fraud-detection models are evaluated on the **same 1,000 transactions** (100 fraud, 900 legitimate).

| Model | Accuracy | Precision | Recall | Flagged cases |
|---|---|---|---|---|
| **A** | **94.5%** | **0.88** | 0.52 | 59 |
| **B** | 93.3% | 0.62 | **0.84** | 135 |
| **C** | 94.0% | 0.70 | 0.70 | 100 |

## Questions

1. Which model would you choose for **fraud screening**, where missing fraud is costly?
2. Which model would suit a **limited manual review team**?
3. Which model has the most **balanced precision and recall**?
4. Why is the model with the highest accuracy not automatically best?

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## Questions

1. Which model would you choose for **fraud screening**, where missing fraud is costly?
2. Which model would suit a **limited manual review team**?
3. Which model has the most **balanced precision and recall**?
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**Possible decisions:**

- **B:** Highest recall (**84%**); detects the most fraud, but flags more transactions.
- **A:** Highest precision (**88%**) and fewest flagged cases; suitable when review capacity is limited.
- **C:** Equal precision and recall (**70%**), offering a balance between them.

**Lesson:** Model selection depends on **missed-case costs, false alarms, and available resources**, not accuracy alone.

# Mini-Quiz

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# Check your understanding

1. Why should the final test set not be used for **hyperparameter tuning**?
2. What is the difference between precision and recall?
3. Why can accuracy be misleading for imbalanced data?
4. When is RMSE more informative than MAE?
5. What does $R^2 < 0$ indicate?
6. What usually happens to recall when the classification threshold is lowered?
7. Why can a model with lower accuracy still be better in a real application?
8. What is the purpose of cross-validation?

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**Suggested answers:** to keep final evaluation unbiased; precision measures correctness among predicted positives, recall measures coverage of actual positives; majority-class predictions can dominate accuracy; RMSE emphasizes large errors; worse than predicting the mean; recall usually increases; error costs may matter more; cross-validation gives more stable validation estimates.

# Lecture 5 Summary

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# Key ideas to retain

## Evaluation foundations

- Training performance is not enough.
- Validation data guides model selection.
- Test data estimates final performance.
- Baselines provide important reference points.
- Leakage can make evaluation unrealistically optimistic.
- Cross-validation gives more stable model-selection evidence.

## Metrics and decisions

- **Regression metrics** measure different error behaviors.
- Confusion matrices reveal types of classification errors.
- Precision, recall, specificity, and $F_1$ answer different questions.
- Imbalanced data requires more than accuracy.
- Thresholds control error trade-offs.
- The best model depends on real-world costs and context.

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# End-of-Lecture Check

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# Five questions to test your understanding

1. What is the difference between validation performance and test performance?
2. Why is a baseline necessary?
3. Which metric would you emphasize when **false negatives** are very costly?
4. How can lowering a threshold improve recall but reduce precision?
5. Why can cost-sensitive evaluation select a model with lower accuracy?

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💡 **Preparation for next lecture:** Review entropy, information gain, categorical splits, Euclidean distance, and why feature scaling matters for distance-based methods.

# Suggested Reading

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# Primary and supporting references

## Primary reference

**G. James, D. Witten, T. Hastie, and R. Tibshirani,** *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Review sections on assessment of model accuracy, validation, cross-validation, classification metrics, and model selection.

## Supporting references

**A. Géron,** *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review classification evaluation, confusion matrices, precision/recall, ROC curves, cross-validation, and model tuning.

**K. P. Murphy,** *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected sections for deeper views of predictive uncertainty, losses, and decision-theoretic evaluation.

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💡 When reading, focus on why a metric is appropriate for a particular decision, not only how to calculate it.